

TRANSFORMERS

AI506 — Advanced Machine Learning

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- Language modeling **objective(s)**
- The **transformer architecture**
- **Attention, Multi-head -attention, position embeddings, attention masks**

Goal: Pick the best next word given that,

$$p(x_t \mid x_{t-1}, x_{t-2}, \dots, x_0)$$

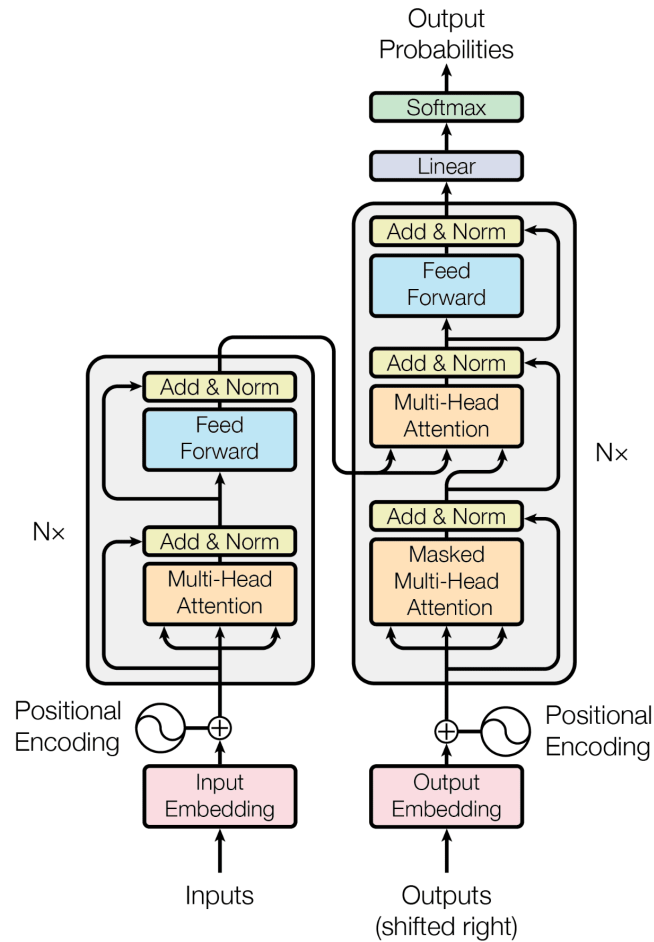
- Pre-training (PT)

$$\mathcal{L}_{\text{PT}} = -\mathbb{E}_{x \sim \mathcal{D}} \sum_t \log p(x_t \mid x_{<t})$$

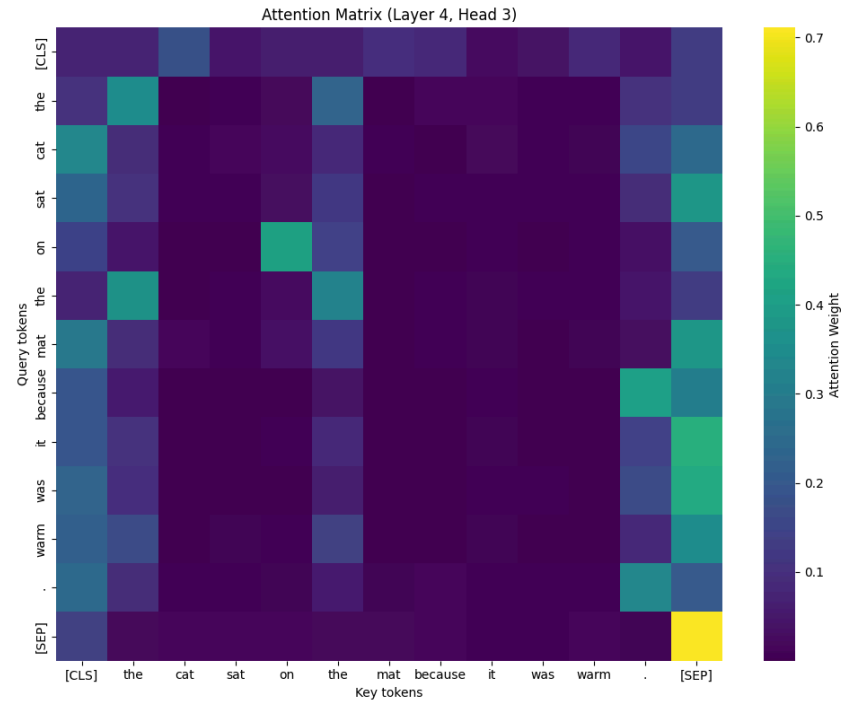
- Supervised fine-tuning (SFT)

$$\mathcal{L}_{\text{SFT}} = -\mathbb{E}_{(x,y) \sim \mathcal{D}_{\text{SFT}}} \sum_t \log p(y_t \mid y_{<t}, x)$$

- RL from human feedback



Attention



$$\text{ATN}(X) = \text{softmax}\left(\frac{Q^T K}{\sqrt{d_k}}\right) V$$

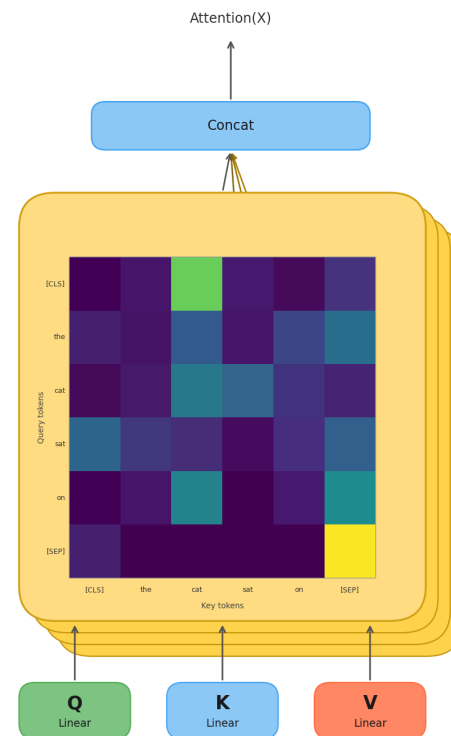
$$Q = XW_Q \quad K = XW_K \quad V = XW_V$$

$$\text{ATN}(X, \mathbf{Z}) = \text{softmax}\left(\frac{Q^T K}{\sqrt{d_k}}\right) V$$

$$Q = XW_Q \quad K = \mathbf{Z}W_K \quad V = \mathbf{Z}W_V$$

$$\text{MultiHead}(X) = \text{head}_1 \# \dots \# \text{head}_h$$

$$\text{head}_i = \text{softmax}\left(\frac{Q^T K}{\sqrt{d_k}}\right) V$$



- Absolute position

$$v_i = W_v \cdot (x_i + p_i)$$

- Relative position

$$q_m^T k_n = x_m^T W_Q^T W_K x_n + p_m^T U_Q^T U_K p_n + b_{n-m}$$

		"Dream" x(1)	"big" x(2)	"and" x(3)	"work" x(4)	"for" x(5)	"it" x(6)
"Dream"	x(1)	α_{11}	α_{12}	α_{13}	α_{14}	α_{15}	α_{16}
"big"	x(2)	α_{21}	α_{22}	α_{23}	α_{24}	α_{25}	α_{26}
"and"	x(3)	α_{31}	α_{32}	α_{33}	α_{34}	α_{35}	α_{36}
"work"	x(4)	α_{41}	α_{42}	α_{43}	α_{44}	α_{45}	α_{46}
"for"	x(5)	α_{51}	α_{52}	α_{53}	α_{54}	α_{55}	α_{56}
"it"	x(6)	α_{61}	α_{62}	α_{63}	α_{64}	α_{65}	α_{66}